Machine Learning a.y. 22-23

Homework 2: Report

Davide Gabrielli - 1883616
gabrielli.1883616@studenti.uniroma1.it

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1 Introduction

For this homework we have been asked to solve a image classification problem choosing a dataset having at least 10 classes and 150 images for each class. In this report I will describe the dataset I have chosen and two different neural networks that I have implemented to solve the problem.

1.1 Dataset

The dataset I have chosen is the 10 Monkey Species dataset from Kaggle. It is composed of 10 different classes of monkeys, each class has more than 150 images as shown in Figure 1. There are almost 1400 images with resolution of 400x300 px or larger and the images are in RGB format.

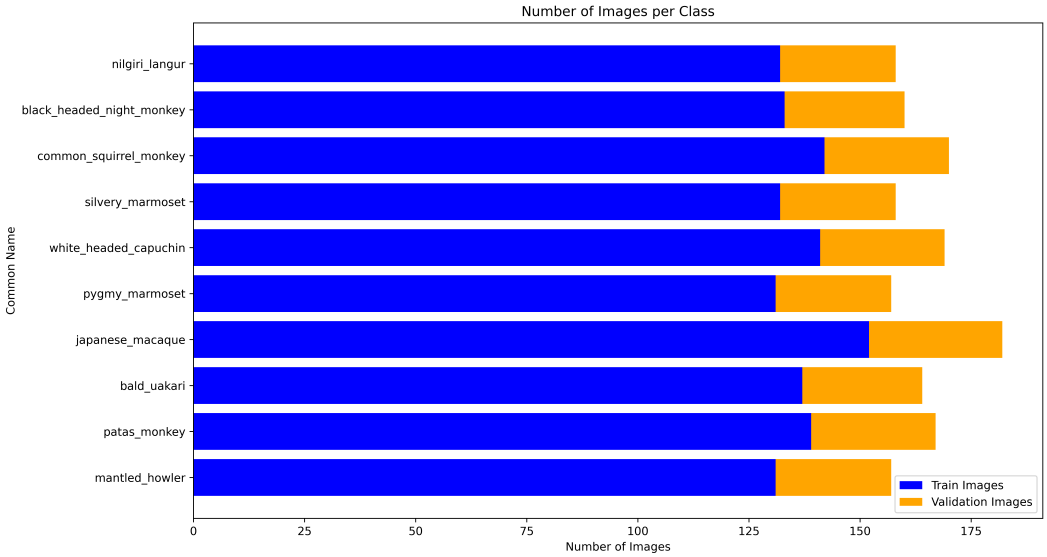

Figure 1: The dataset is composed of 10 different classes of monkeys, each class has ¿150 images.

2 Preprocessing

In this section I will describe the preprocessing steps I have applied to the dataset.

2.1 Normalization

The first step I have applied to the dataset is the normalization of the images. Since the image pixels are in the range [0, 255] I have normalized them to the range [0, 1] dividing each pixel by 255. This step is important because it allows the network to converge faster and it is also important to avoid numerical issues.

2.2 Dataset splitting

Since the dataset already have a train and a validation set, I loaded them separately and used them to train and evaluate the models.

2.3 Data augmentation

The second step I have applied to the dataset is the data augmentation. I have used the ImageDataGenerator class from TensorFlow which allows to apply different transformations to the images in real time. So the following transformations have been applied to the images:

- **Rotation**: the images have been rotated by a random angle between -20 and 20 degrees.
- **Width shift**: the images have been shifted horizontally by a random amount between -0.2 and 0.2 of the width.
- **Height shift**: the images have been shifted vertically by a random amount between -0.2 and 0.2 of the height.
- **Shear**: the images have been sheared by a random angle between -20 and 20 degrees.
- **Zoom**: the images have been zoomed by a random amount between 0.8 and 1.2.
- **Horizontal flip**: the images have been flipped horizontally with a probability of 0.5.

3 Convolutional Neural Network

In this section I will describe

3.1 Training

3.1.1 Hyperparameter tuning

3.2 Evaluation

4 Autoencoder

In this section I will describe

4.1 Training

4.1.1 Hyperparameter tuning

4.2 Evaluation

5 Conclusions